

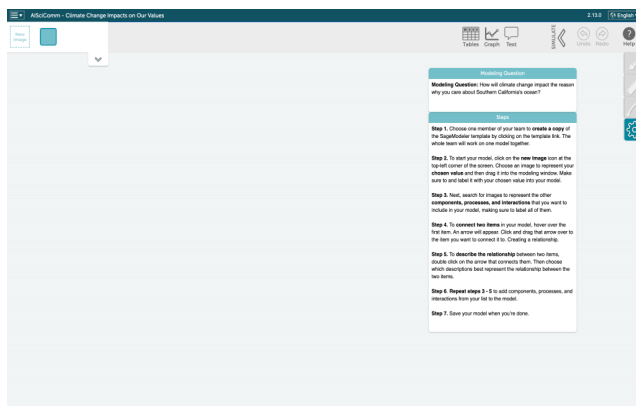
Module 2.1: Building a Model

SageModeler Instruction

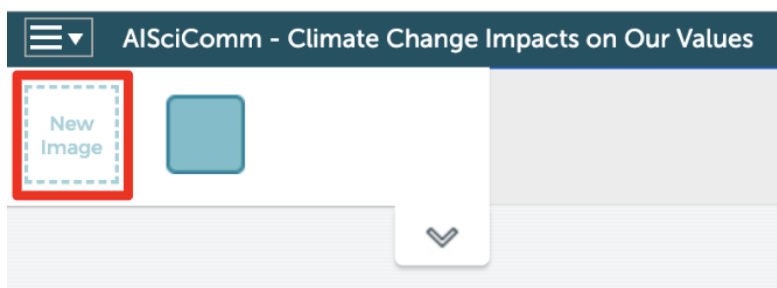
Follow these steps with your team to create your model on how climate change impacts your chosen value.

1. Choose one member of your team to **create a copy** of the [SageModeler Template](https://tinyurl.com/AISciModel) or visit <https://tinyurl.com/AISciModel>.

The whole team will work on one model together.



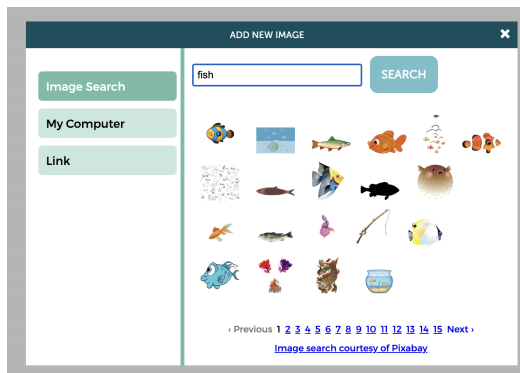
2. To start your model, click on the new image icon at the top-left corner of the screen.



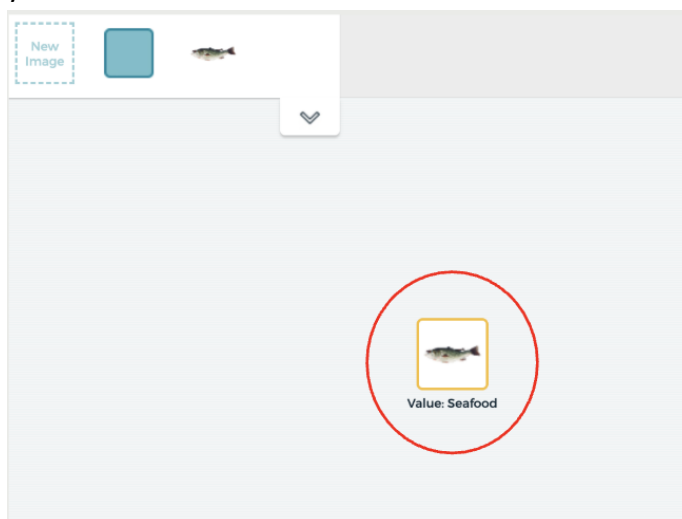
Choose an image to represent your chosen value. Press “Add Image.”

Example: Search “fish” for the value seafood.

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Drag the image in the modeling window. Make sure to **label** it with your **chosen value** in your model.



3. Search for images to represent the other **components, processes, and interactions** that you want to include in your model. **Label all of them.**

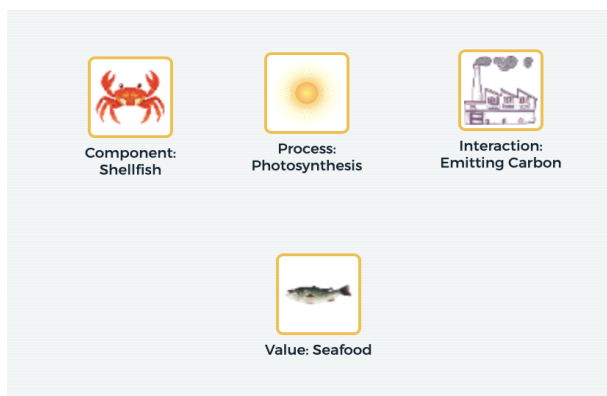
Example:



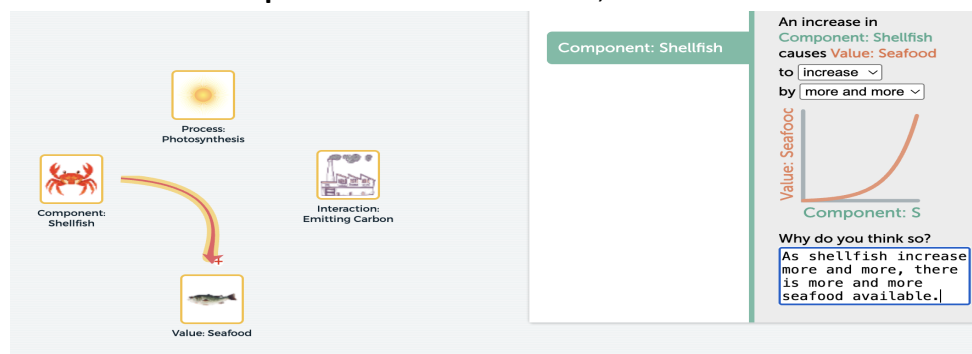
4. To **connect two items** in your model, hover over the first item. An arrow will appear.



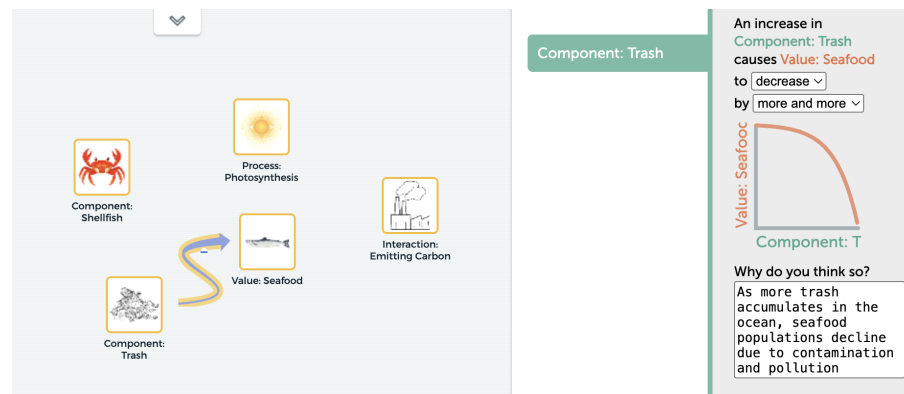
Click and drag that arrow over the item you want to connect it to. Creating a causal relationship.



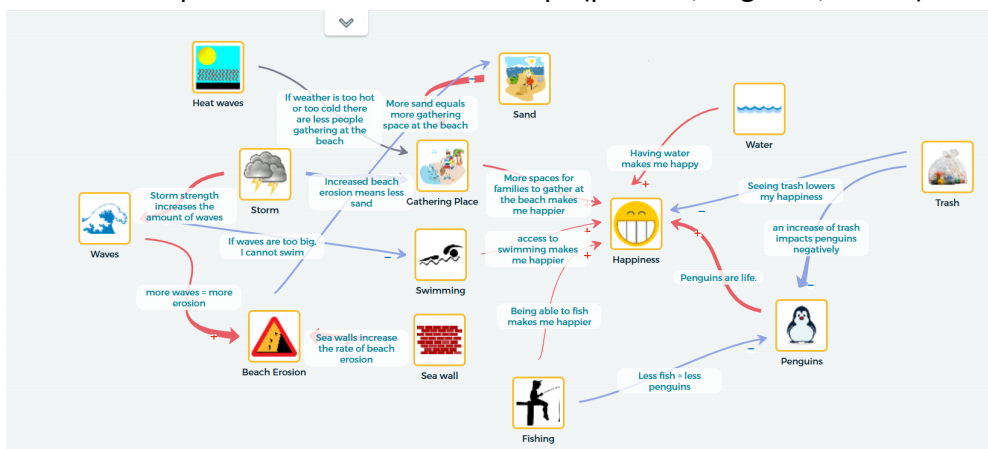
5. To **describe the relationship** between two items, double-click on the arrow that connects them.
 - a. **Positive relationship:** As one factor increases, the second factor also increases.



- b. **Negative relationship:** As one factor increases, the second factor decreases.



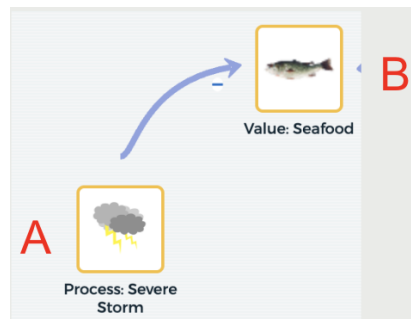
- c. **Varied relationship:** As one factor increases, the second factor can increase or decrease depending on different conditions.
- d. **OPTIONAL:** To make sense of the various interactions, you may add a brief explanation for the relationships (positive, negative, varied).



6. There are different types of causal relationships that you can model. Try to model different types of causal relationships.
 - a. **Linear Chain (A → B)**
Concept: A direct cause-and-effect relationship where one element or process directly impacts another.

Example:

A heavy storm (A) physically prevents fishing vessels from leaving port, directly reducing the immediate supply of fresh seafood (B).

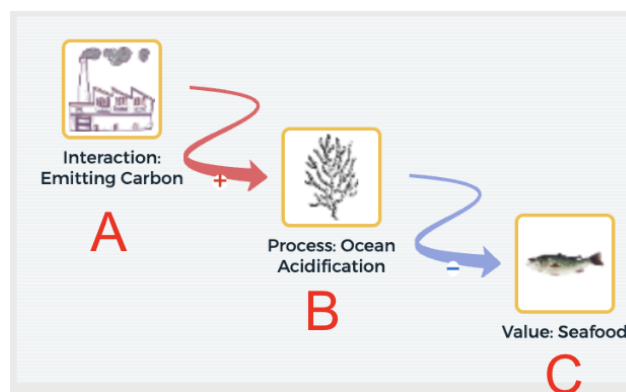


b. **Causal Chain (A → B → C)**

Concept: A multi-step sequence where an initial event triggers an intermediate change, which then leads to a final outcome.

Example:

When factories release carbon dioxide into the air (A), the ocean absorbs it and becomes acidic (B), dissolving oyster shells and reducing seafood yields for farmers (C).

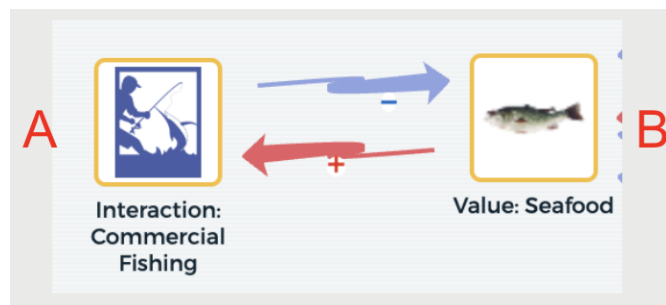


c. **Feedback Loop (A → B; B → A)**

Concept: A circular relationship where the outcome of a process feeds back into the original cause, amplifying (positive feedback) or reducing (negative feedback) the effect over time.

Example:

Intense fishing (A) drives down wild fish populations (B), creating market scarcity that raises prices and incentivizes companies to fish even harder (A), driving stocks into a deeper decline.



d. **Indirect Link ($A \rightarrow B$; $B \rightarrow C$; $A \rightarrow C$)**

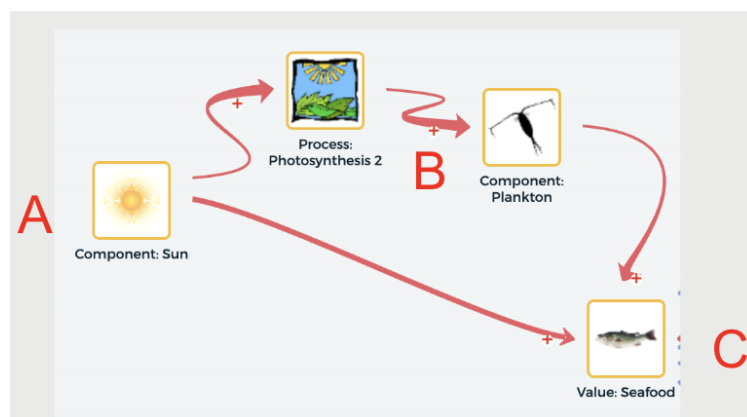
Concept: A variable (A) influences an outcome (C) both indirectly through an intermediate variable (B) and through a separate direct pathway.

Example:

Indirect Pathway ($A \rightarrow B$; $B \rightarrow C$): Increased sunlight boosts phytoplankton growth

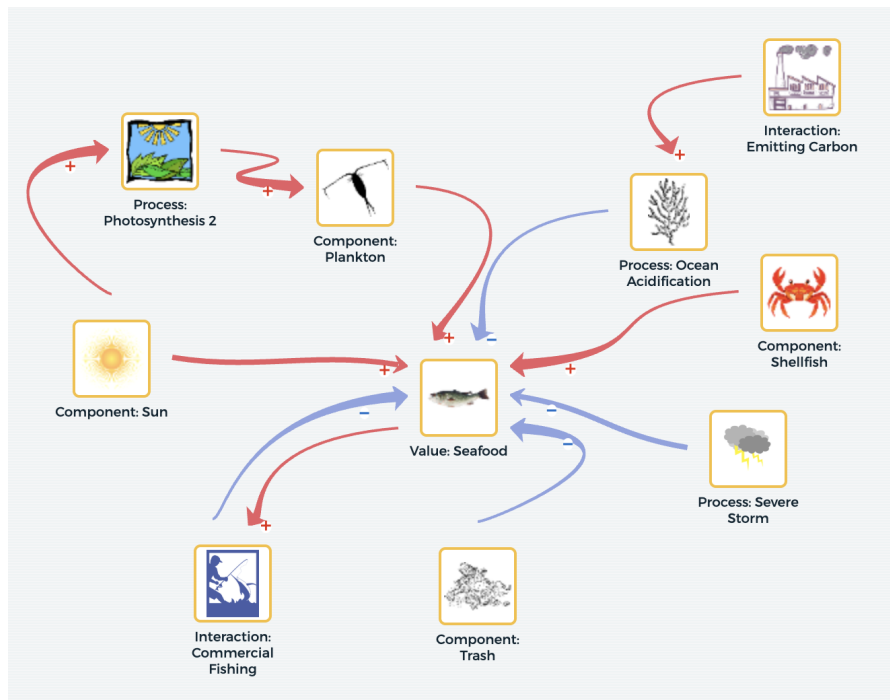
($A \rightarrow B$), which feeds small marine life and ultimately builds up commercial fish populations ($B \rightarrow C$).

Direct Pathway ($A \rightarrow C$): Sunlight directly warms surface waters and alters fish migration routes ($A \rightarrow C$), changing where fish gather and affecting local seafood availability and prices.

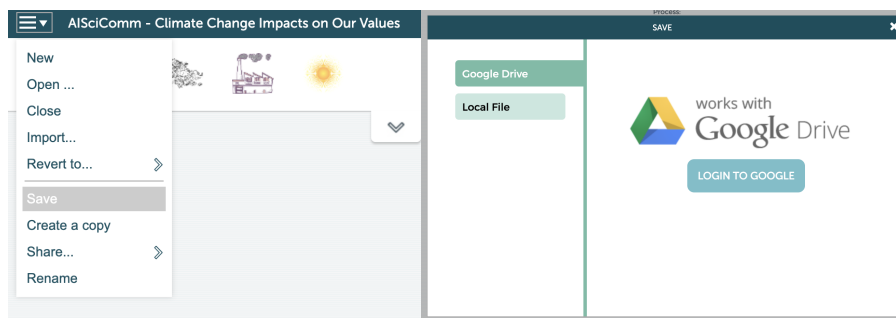


- Continue to add components, processes, and interactions from your list to the model. Create different causal relationships- linear chains, causal chains, feedback loops, and indirect links.

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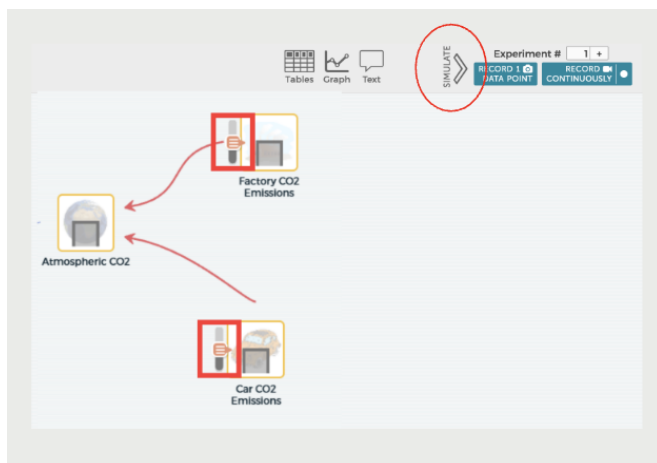
8. Save your model when you're done. Connect to Google Drive to save your model in a shareable location.



9. Now that your team has a complete model, take a moment to simulate change to see how changes in a portion of your model impact your chosen value.

Click on the **Simulate** icon in the upper right-hand corner of SageModeler to simulate change on a different factor within your model.

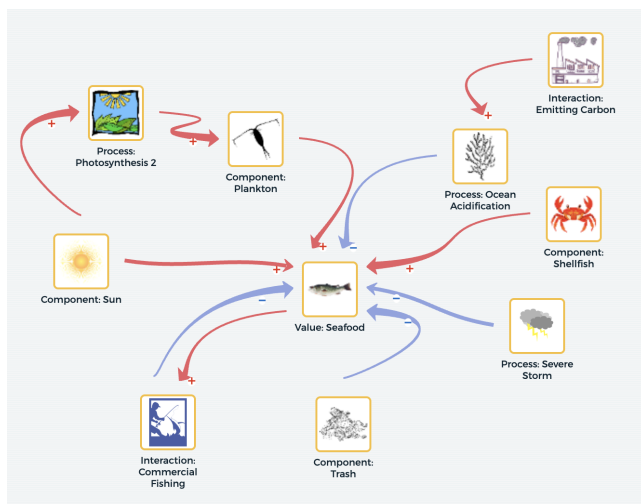
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As you test your simulations, think about the following:

How does a change in a component, process, or interaction within your model affect your team's chosen value?

10. Now that your team has completed their model, **take a screenshot and save your chatbot conversation** in your Google Drive. Save this snapshot as you will need it for Module 4.4.



- a. Devices will differ in ways to take a snapshot.
 - i. If on a Windows computer
 1. Press **Window + Shift + S** at the same time
 - ii. If on a Mac computer
 1. Press **Command (⌘) + Shift + 4** at the same time
 - iii. If on a Chromebook
 1. Press **Ctrl + Shift + Show Windows** at the same time